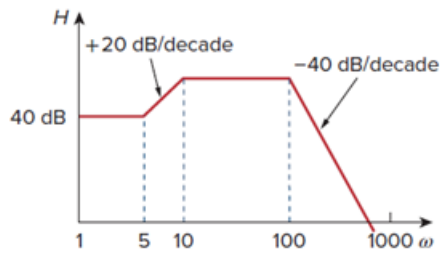


Problem

Obtain the transfer function $H(\omega)$ corresponding to the Bode plot in Fig. 14.20.

Figure 14.20:



Step-by-step solution

Step 1 of 3

Refer to Figure 14.20 from the textbook

To obtain $\mathbf{H}(\omega)$ from the bode plot; we keep in mind that a zero always causes an upward turn at a corner frequency, while a pole causes a downward turn. We notice from the figure 14.20 that there is a straight line shifted from origin to 40 dB indicates that there is a 40dB gain;

Thus,

$$40 = 20 \log_{10} K$$

$$\log_{10} K = \frac{40}{20}$$

Rewrite the equation.

$$\begin{aligned} K &= 10^{\frac{40}{20}} \\ &= 10^2 \\ &= 100 \end{aligned}$$

Step 2 of 3

In addition to the 40 dB gain, we notice that there are three factors with corner frequencies at $\omega = 5$, 10 and 100 rad/s. Thus,

1. A zero at $z = 5$ with slope 20 dB/decade to cause a upward turn. The zero at $z = 5$ is determined as $\left(1 + \frac{j\omega}{5}\right)$.

2. A pole at $p = 10$ with slope -20 dB/decade to cause a straight line and counteract the pole at 10. The pole is determined as $\frac{1}{\left(1 + \frac{j\omega}{10}\right)}$.

3. Another pole at $p = 100$ with slope -40 dB/decade to cause a downward turn. The pole at $p = 100$ is $\frac{1}{\left(1 + \frac{j\omega}{100}\right)^2}$.

Step 3 of 3

Putting all these together gives the corresponding transfer function as

$$\begin{aligned}
 \mathbf{H}(\omega) &= \frac{100\left(1 + \frac{j\omega}{5}\right)}{\left(1 + \frac{j\omega}{10}\right)\left(1 + \frac{j\omega}{100}\right)^2} \\
 &= \frac{100\left(\frac{5 + j\omega}{5}\right)}{\left(\frac{10 + j\omega}{10}\right)\left(\frac{100 + j\omega}{100}\right)^2} \\
 &= \frac{(100)(10)(10000)(5 + j\omega)}{5(10 + j\omega)(100 + j\omega)^2} \\
 &= \frac{2,000,000(j\omega + 5)}{(j\omega + 10)(j\omega + 100)^2}
 \end{aligned}$$

Substitute $j\omega$ for s .

$$\mathbf{H}(\omega) = \frac{2,000,000(s + 5)}{(s + 10)(s + 100)^2}$$

Therefore, the transfer function $\mathbf{H}(\omega)$ is $\boxed{\frac{2000,000(s + 5)}{(s + 10)(s + 100)^2}}$.